CURRENT ELECTRICITY

LEVEL 1 – NCERT & Board Pattern (150 Questions)

SECTION 1: ELECTRIC CURRENT & CHARGE FLOW (Q1–20)

1. Define electric current. What is its SI unit?
2. State the relationship between current, charge, and time.
3. How many electrons are required to constitute 1 coulomb of charge?
4. Distinguish between conventional current and electron flow.
5. What do you mean by the direction of electric current?
6. A charge of 96,500 C passes through a conductor in 1 hour. Calculate the current.
7. How much charge flows through a conductor when a current of 2 A flows for 10 minutes?
8. Can the direction of current be opposite to the direction of electron flow? Explain.
9. State the principle of conservation of charge.
10. What is the physical meaning of 1 ampere?
11. Distinguish between DC and AC current.
12. In a conductor, how are electrons arranged?
13. What is the effect of temperature on the motion of free electrons in a conductor?
14. Explain the concept of current density.
15. How does the current density relate to the electric field?
16. What is the relationship between current and number density of free electrons?
17. A wire carries a current of 5 A. How many electrons pass through a cross-section in 2 seconds?
18. What is meant by steady-state current?
19. How does the current vary with time in a DC circuit?
20. Explain why current is considered to flow from positive to negative terminal.

SECTION 2: DRIFT VELOCITY & MOBILITY (Q21–45)

21. Define drift velocity.
22. Distinguish between random motion and drift velocity of electrons.
23. What is the typical value of drift velocity in a conductor?
24. Derive the relationship: $I = nAe\,v_d$.
25. Define mobility of charge carriers.
26. What is the SI unit of mobility?
27. State the relationship between drift velocity and electric field.
28. How does mobility depend on temperature?
29. Why is the drift velocity very small compared to thermal velocity?
30. In a conductor, if the electric field is doubled, what happens to drift velocity?
31. Explain the relaxation time of an electron.

32. What is the relationship between relaxation time and mobility?
33. In a copper wire, what factors affect the drift velocity of electrons?
34. How does drift velocity depend on the cross-sectional area of the conductor?
35. Calculate drift velocity if $n = 8.5 \times 10^{28}\ \mathrm{m}^{-3}, I = 10\ \mathrm{A}, A = 10^{-6}\ \mathrm{m}^2$.
36. Why do we need to apply an external electric field to produce current?
37. What is the relationship between current and electric field?
38. Explain why drift velocity increases with applied voltage.
39. How does the number density of free electrons affect current?
40. In two wires of same material but different cross-sections, which has greater drift velocity for the same current?
41. Define relaxation time and explain its significance.
42. How is mobility related to the mean free path of electrons?
43. Why is mobility of electrons greater in semiconductors than in metals?
44. What happens to mobility when impurities are introduced in a conductor?
45. Explain the concept of mean free path of an electron.

SECTION 3: RESISTANCE & RESISTIVITY (Q46–85)

46. Define resistance. State its SI unit.
47. State Ohm's law. Is it a universal law? Explain.
48. Define resistivity and state its SI unit.
49. Derive the relationship: $R = \rho L/A$.
50. How does resistance depend on the length of a conductor?
51. How does resistance depend on the cross-sectional area of a conductor?
52. What is the difference between resistance and resistivity?
53. Why does resistivity depend on material but resistance depends on both material and geometry?
54. How does the resistivity of metals vary with temperature?
55. How does the resistivity of semiconductors vary with temperature?
56. Define temperature coefficient of resistance.
57. What is the significance of a negative temperature coefficient?
58. A copper wire of length L and cross-section A has resistance R. What will be the resistance of a wire of the same material with length $2L$ and cross-section $2A$?
59. Compare the resistivity of copper and nichrome.
60. Why are some materials ohmic while others are non-ohmic?
61. A wire is stretched to double its length. How is its resistance affected?
62. Two wires of copper and iron of the same length and area are connected in series. Which wire becomes hotter?
63. Calculate the resistance of a copper wire of length 100 m and area $1\ \mathrm{mm}^2$ ($\rho = 1.7 \times 10^{-8}\ \Omega\,\mathrm{m}$).
64. What is the effect of temperature on the resistivity of a superconductor?

65. Explain the concept of conductivity and conductance.
66. How are conductivity and resistivity related?
67. What is the SI unit of conductance?
68. A conducting wire is cut into two equal parts. What is the ratio of their resistances?
69. Why does a thin wire have greater resistance than a thick wire of the same material and length?
70. What happens to the resistance of a conductor when it is bent?
71. Explain why the filament of an electric bulb becomes very hot but connecting wires do not.
72. Define specific resistance or resistivity with examples.
73. How does resistivity depend on the atomic structure of a material?
74. What is meant by resistance in series and parallel?
75. In a metal, as temperature increases, does resistivity increase or decrease?
76. In a semiconductor, as temperature increases, does resistivity increase or decrease?
77. Explain the microscopic origin of resistance.
78. What is the relationship between resistivity and relaxation time?
79. How does resistivity change when a conductor undergoes strain?
80. Why do we need to know the resistivity of materials?
81. In what way does Ohm's law fail in non-ohmic devices?
82. Calculate resistivity if $R = 10\ \Omega$, $L = 2\ \mathrm{m}$, $A = 0.5\ \mathrm{mm}^2$.
83. A metal has temperature coefficient $0.005/°\mathrm{C}$. If its resistance at $0°\mathrm{C}$ is $100\ \Omega$, find resistance at $100°\mathrm{C}$.
84. Why are alloys like nichrome used in heating elements?
85. What is the difference between resistivity of conductors, semiconductors, and insulators?

SECTION 4: EMF & INTERNAL RESISTANCE (Q86–110)

86. Define EMF (electromotive force).
87. What is the SI unit of EMF?
88. Distinguish between EMF and potential difference.
89. What is internal resistance? Why does every battery have internal resistance?
90. State the relationship between terminal voltage, EMF, and internal resistance.
91. When is the terminal voltage equal to EMF?
92. When is the terminal voltage less than EMF?
93. A battery of EMF 12 V and internal resistance $0.5\ \Omega$ is connected to a load of $11.5\ \Omega$. Find current and terminal voltage.
94. What happens to the terminal voltage when current drawn increases?
95. Explain why a cell can supply only limited current.
96. What is the difference between an ideal and a real cell?
97. How does internal resistance affect efficiency of a battery?
98. Derive $V = E - Ir$.
99. When is power dissipated in internal resistance maximum?

100. How can internal resistance of a battery be determined experimentally?
101. What is meant by the loading effect of internal resistance?
102. Explain why maximum power is delivered when $R = r$.
103. Why does a battery become warm during discharge?
104. What happens to terminal voltage while charging a battery?
105. Explain back EMF in a charging cell.
106. Can internal resistance of a battery be zero? Explain.
107. How does temperature affect internal resistance?
108. In a circuit with multiple batteries, how is net EMF calculated?
109. How does EMF of a cell differ during charging and discharging?
110. Why does terminal voltage decrease when more devices are connected in parallel?

SECTION 5: OHM'S LAW & KIRCHHOFF'S LAWS (Q111–130)

111. State Ohm’s law (macroscopic form).
112. State microscopic form of Ohm’s law.
113. What is the condition for validity of Ohm’s law?
114. What are ohmic and non-ohmic devices? Give examples.
115. Draw a V–I graph for an ohmic resistor.
116. Draw a V–I graph for a non-ohmic device (e.g., diode).
117. State Kirchhoff’s current law.
118. State Kirchhoff’s voltage law.
119. What physical principle underlies Kirchhoff’s current law?
120. What physical principle underlies Kirchhoff’s voltage law?
121. In a series circuit: V_1 = 2 V, V_2 = 3 V, total EMF = 6 V. Find V_3.
122. Use Kirchhoff’s laws to find current in each branch of a network.
123. How many independent equations can be written in a circuit with n loops?
124. Can Kirchhoff’s laws be applied to AC circuits?
125. Can sum of potential rises equal sum of potential drops in a loop? Explain.
126. What is the significance of Kirchhoff’s laws?
127. How can Kirchhoff’s laws be used to find equivalent resistance?
128. Explain why Kirchhoff’s laws are based on conservation laws.
129. How do we choose current direction in multi-loop circuits?
130. Can Kirchhoff’s voltage law be applied when magnetic field is time-varying? Explain.

SECTION 6: SERIES & PARALLEL COMBINATIONS (Q131–150)

131. What is meant by resistors in series? Draw a diagram.
132. Derive formula for equivalent resistance in series.
133. In series combination, does current remain same?
134. In series combination, does voltage vary?

135. What is the advantage of series combination?
136. What is the disadvantage of series combination?
137. What are resistors in parallel? Draw a diagram.
138. Derive formula for equivalent resistance in parallel.
139. In parallel combination, does current remain same?
140. In parallel combination, does voltage remain same?
141. What is the advantage of parallel combination?
142. What is the disadvantage of parallel combination?
143. Three resistors 2 Ω, 3 Ω, 6 Ω are in parallel. Find their equivalent resistance.
144. Three resistors 2 Ω, 3 Ω, 6 Ω are in series. Find total resistance.
145. Explain current divider rule with derivation.
146. Explain voltage divider rule with derivation.
147. In a parallel circuit, if one resistor is zero, what happens?
148. If a series resistor becomes very large, what happens to current?
149. Can series and parallel resistors exist in one circuit?
150. How is total resistance calculated in a mixed combination?

Total Questions: 150
Difficulty Level: NCERT & 12th Board Pattern
Coverage: All major concepts of Current Electricity